

REBECCA DRIVER

A post-graduate student working in computational chemistry with a focus on developing sustainable materials.

EDUCATION

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| 2025 - Present | MPhil/PhD in Chemistry Research
King's College London <ul style="list-style-type: none">Developing multiscale models for complex biobased fluids using molecular simulations and machine learning models, for a more sustainable future |
| 2024 - 2025 | MSci Natural Sciences Tripos (Chemistry)
University of Cambridge
Class: 1st <ul style="list-style-type: none">Specialising in theoretical and computational chemistryCourses include: Machine Learning for Atomistic Simulations, DFT and molecular simulations |
| 2021 - 2024 | BA (Hons) Natural Sciences Tripos (Chemistry)
University of Cambridge
Class: 2.1 <ul style="list-style-type: none"><i>Year 1:</i> Physics, Mathematics, Chemistry, Materials Science<i>Year 2:</i> Chemistry and History & Philosophy of Science<i>Year 3:</i> Advanced Physical & Theoretical Chemistry (Statistical Mechanics, Perturbation Theory, and related topics) |

SKILLS

- Experienced in Python for large-scale data analysis and visualisation, with a strong foundation in molecular simulations and DFT.
- Natural Sciences training and multidisciplinary research experience give a broad interdisciplinary perspective in the sciences.

EXPERIENCE

2025 - Present: Postgraduate Research

Dr Francisco Martin-Martinez

Multiscale computational modelling of biobased fluids for novel applications

- Developing multiscale models for bio-oils and complex organic liquids using molecular dynamics, DFT calculations, and machine-learning approaches.
- Building techniques to study structure–property relationships, to support the design of next-generation biobased materials.

2024-25: Part III (MSci) Project

Prof. Rosanna Collepardo-Guevara and Dr Kieran Russell

Exploring liquid - liquid phase separation of multi-domain proteins

- Performed molecular simulations using the group's coarse-grained model, OpenCGChromatin, to study residue-specific interactions governing LLPS in a large multi-domain protein.
- Gained experience in coarse-graining, advanced sampling, and further experience visualising scientific data.
- Received a high first (80.5 %) for this project.

Summer 2024: Summer Research Project

Dr Stephen Cox and Anna Bui

Studying the relationship between solvation forces and lateral pressure using molecular simulations

- Conducted an 8-week funded project (Cambridge Opportunity Fund) investigating how solvation forces contribute to lateral pressure in confined liquids using molecular simulations (MD and MC).

Academic and Pastoral Tutoring

2021 - 2025

- Served as **Selwyn College JCR Access Officer (2022–23)**, organising outreach events, teaching sessions, and presentations for prospective applicants.
 - Acted as a **STEM SMART pastoral mentor (2023–24)**, supporting widening participation in Cambridge undergraduate admissions.
 - Provide **GCSE and A-level mathematics tutoring**, designing tailored lesson plans and monitoring student progress.
 - **Graduate teaching assistant** at King's College London, teaching undergraduate students about computational and quantum chemistry.
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